

relies heavily on logging or outputs on serial ports using either a monitor in the ROM or using a debug server like the Remedy Debugger. Note that this is not what we call debugging in our usual sense, the programmer cannot step through the code line by line, rather the output of each line of execution is printed out to the serial port or a flash memory, the user then has to manually go through the code results to find out what exactly is going wrong. Another example of an external debugger would be the ADB (Android Debugging Bridge) used in Android smartphones. You start an ADB server on your host and connect to the target i.e. the phone using a USB-cable to the system. You can then debug the code on your smartphone usually an application from the host. Hackers utilise ADB to inject exploits into the phone which help bypass the restrictions on the bootloader and root the phone.

In Circuit Emulators

An In Circuit emulator is a speciality hardware device, it is developed with what is known as a bond-out processor. These processors have internal signals and buses brought out to external pins for monitoring. These signals provide the developer with the information about the current state of the processor. An ICE replaces the target chip and acts like



RTE870_C In circuit emulator

the processor to the rest of the circuit but it comes with a host of internal mechanism which allow the developer to see what's going on inside, set breakpoints, load new code, grab traces etc.

The problem is the cost associated with this kind of emulator is the cost of manufacturing a bond-out version of the target chip, for the current processors the speed have increased so much that it is nigh impossible to take anything off the chip. Another problem is that the ICE requires the chip (i.e. the ICE chip) to be in socket, or requires a special adapter mounted in the target chip socket so that the ICE can connect to it's lines.

For this particular reason recently JTAG (Joint Test Action Group) based hardware debuggers are being considered to be ICEs. This JTAG based hardware provides access equivalent to ICEs using on-chip hardware even